

Mathematics A

Unit 2 • Chapter OL-T16 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

2 classified items • 7 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

Dr Eslam Ahmed

Elite IGCSE Mathematics • eliteigcse.com

WhatsApp +20 112 000 9622

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Contents

Unit 2 • Expertise • Unit 2 • Chapter OL-T16 • Expertise Q16+ • 2 items • 7 marks

Chapter	Items	Marks	Starts
01. OL-T16 Algebra Toolkit	2	7	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Algebra Toolkit

Unit 2 • Expertise • 7 marks • 2 item(s)

June 2024 | 4MA1-2H Q20 | 3 marks

Signed, decimal or fractional formula substitution

June 2023 | 4MA1-2H Q17 | 4 marks

Subject appears in two linear terms

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

4MA1-2H Q20

3 marks

20 Given that $k = x - y$ and $x = \frac{1}{4y}$

express $\frac{5k}{x+2}$ in the form $\frac{a-by^2}{c+dy}$ where a, b, c and d are integers.

WORKING SPACE

4MA1-2H Q20

3 marks

20 Given that $k = x - y$ and $x = \frac{1}{4y}$
express $\frac{5k}{x+2}$ in the form $\frac{a-by^2}{c+dy}$ where a, b, c and d are integers.

.....

Algebraic substitution

3 marks

Substitute

$$x = 1/(4y), \quad k = 1/(4y) - y.$$

Simplify

$$\frac{5k}{x+2} = \frac{5-20y^2}{1+8y}.$$

Final answer

$$\frac{5-20y^2}{1+8y}$$

4MA1-2H Q17

4 marks

(b) Make e the subject of $w = \sqrt{\frac{e+g}{ef-d}}$

.....
(4)

WORKING SPACE

4MA1-2H Q17

4 marks

(b) Make e the subject of $w = \sqrt{\frac{e+g}{ef-d}}$

.....
(4)

Make e the subject of a square-root formula

4 marks

Square

$$w^2 = \frac{e+g}{ef-d}.$$

Clear the denominator

$$w^2(ef-d) = e+g \Rightarrow w^2ef - w^2d = e+g.$$

Collect e

$$e(w^2f-1) = g+w^2d.$$

Divide

$$e = \frac{g+w^2d}{w^2f-1}.$$

Final answer

$$e = \frac{g+w^2d}{w^2f-1}$$